



LUT
University

CT60A5401

GAME DEVELOPMENT PROJECT

ACADEMIC YEAR 2022-2023

ASSOC. PROF. JUSSI KASURINEN (D.SC.)



TODAY'S AGENDA

- **What is the aim of this course, how this course can be passed.**
- **General topics on how to start making games.**

COURSE OBJECTIVES

GAME DEVELOPMENT PROJECT

INTRODUCTION

Objectives

The objective for this course is for students to learn how to use the selected game development tools, and enable them to independently design and develop a small game program for the supported platforms or work as a part of a team developing a larger game product. After the course, the student is able to do independent work and is capable of acquiring further knowledge concerning the taught game development tool.

INTRODUCTION

Content

- Basics of the game development tool,
- Introduction to game engines and their functions.
- Basics of working with 3D objects,
- Introduction to game development-related programming.
- Basics of sound engineering.
- Gamification and
- Serious games.



- **To give an idea how to develop games with a modern game engine.**
- **Teach introductory stuff about the game engines, game development in general and**
- **Understand how game development differs from software development and traditional programming work.**
- **Understand what different areas of entertainment software engineering there are.**
- **Enable everyone to develop games and be able to participate to the Finnish Game Jam event (a.k.a. at FGJ Lappeenranta.)**
 - FGJ event available as an alternative to complete the course “Code Camp”

TOOLS

- **On this course, the “default” development environment we use in the examples is the Unity 3D –game engine.**
 - However, in your own project you can use whatever game engine you want to.
 - Same goes to animation tools, SFX editors and others.
- **Additionally, you need to create two videos, so recording software and editing tools will be useful.**
- **...And write a game design document, so some form of office tools.**



WHY UNITY?

- **Unity game engine (www.unity3d.com) is probably right now the most used game engine in the world.**
 - It has free standard version, allowing even commercial development. (After 100k profit requires Pro-version.)
 - Easy to learn, wealth of tutorials and stuff in the Internet.
 - Compatible with most game development-related tools.
 - Easy interface, need to do programming work only when necessary.
 - Compatible with every gaming-relevant platform, mobile included.
 - Extendable to fit almost any need or desire for a game development IDE.

Some examples (games made with Unity):



KERBAL SPACE PROGRAM



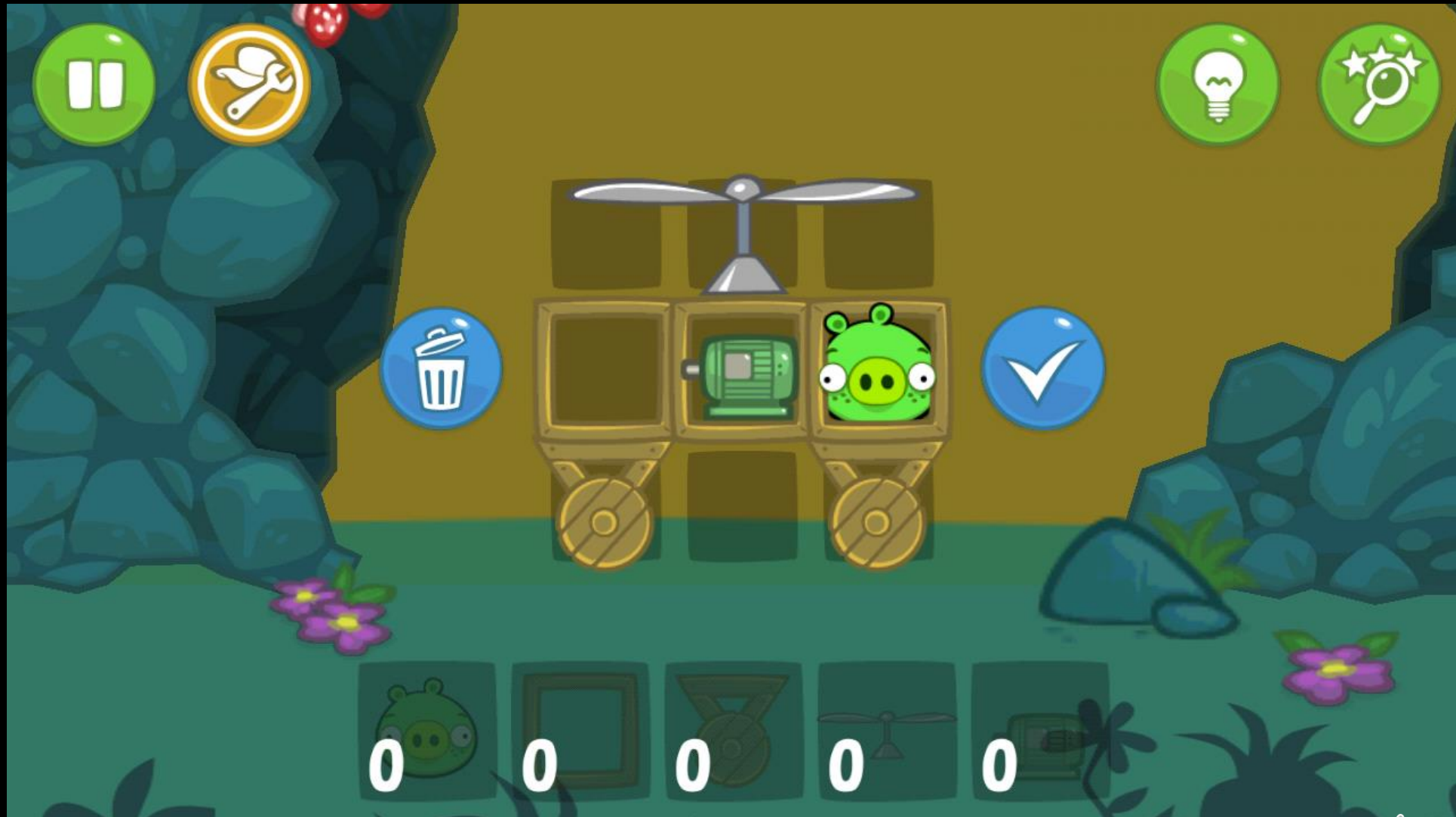
FALL GUYS (MEDIATONIC)



WASTELAND™ 3



BAD PIGGIES (ROVIO)



CONCERNING THE COURSE

...OK so realistically we might not make the next Call of Duty on this course (but we *could*); the idea is that this assisted self-study course enables you to do stuff with Unity and other game engines.

OK. SO WHAT IS "ASSISTED SELF-STUDY"

- It means that this course will not have weekly lectures or demos or exercises, but you can seek help from course teaching staff or by asking from other participants.
- All the materials are available you when you start this course, and you are free to select whenever you wish to complete this during the academic year.
- If you get stuck, go to the Moodle pages discussion forum or email the course assistants.
 - Obviously keep in mind that you are free to select your tools; it also means that you should select tools you are either familiar with, or have enough support systems to help you if you get stuck. If unsure what to do, use Unity.

PLEASE OBSERVE

From 2017 feedback: *” Teaching methods??????
The lectures were horrible and everything that you
needed to know you had to do by yourself ”*

This course is an assisted **SELF STUDY** course!

Meaning:

- You **will** need to learn things by your own.
- You **will not** be taught everything at lectures, or in the exercises, or in general, by the teacher.

(The lectures might still be horrible, though.)



HOW TO PASS

GAME DEVELOPMENT PROJECT



PROJECTS

This course is based around two projects and an exam:

1. Project: Tools and Game design document

- Select your development tools, and create a short 5-10 minute video where you demonstrate how they work in general, and how games are developed with this system. (For example explain how game projects and repositories work with this particular engine)
- Write a Game Design Document for your own game.
- Pass/Fail grading.

2. Own game

- Defined in a game design document of your own design.
- Implement the game, make it available online, that's it!
- Can be something relatively simple, think Breakout, Space war, Pacman, Mario Bros. (the arcade game, not the platformer on NES).
- Additionally, make one 1-2min teaser trailer for your game.
- Grade 0-5 based on implemented features.

FEATURE FORM AND EXAM

- Starting Fall 2022, the project 2 is graded 0-5 based on the amount and difficulty of the features you developed for your game.
 - List of which features are mandatory, and which score you points in the Moodle pages.
- Also starting Fall 2022, instead of peer reviews there will be a Moodle exam, which will be based on the course lectures. Again, graded 0-5.
- You need to pass Project 1, and get at least grade 1 from both the Exam and Project 2 for a passing grade.

GAME DESIGN DOCUMENT

GAME DEVELOPMENT PROJECT



HOW THE GAMES ARE DESIGNED

Brainstorming; lead **game designer** and the team in free form sessions.

Prototyping; cobble up playable test version for the team to judge.

Source for ideas: movies, fairy tales, books, TV, daydreams, twisted **sense of humor**....

“If you cannot summarize your idea in **three slides**, it is a bad idea.”



GAME DESIGN DOCUMENT

As a part of this course, you will be expected to create a short document describing your idea for a game.



GAME DESIGN DOCUMENT

The most important aspects written down:

- General theme, main characters.
- Genre (shooter, platformer, puzzle etc.)
- Mostly important features
- Story, if there is any.
- Technical decisions



GAME DESIGN DOCUMENT: A TEMPLATE

See the template here:

https://docs.google.com/document/d/1zV330cE5tqbuQC0aJgMhEAn8nxzbb_UU0rOl0pCGJmU/edit?usp=sharing

Use this as your starting point!



REMEMBER: THINK SIMPLE!

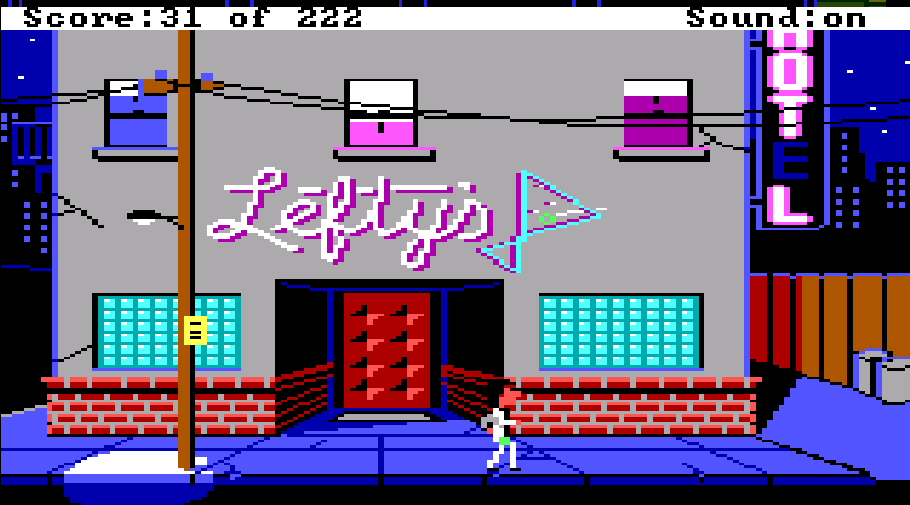
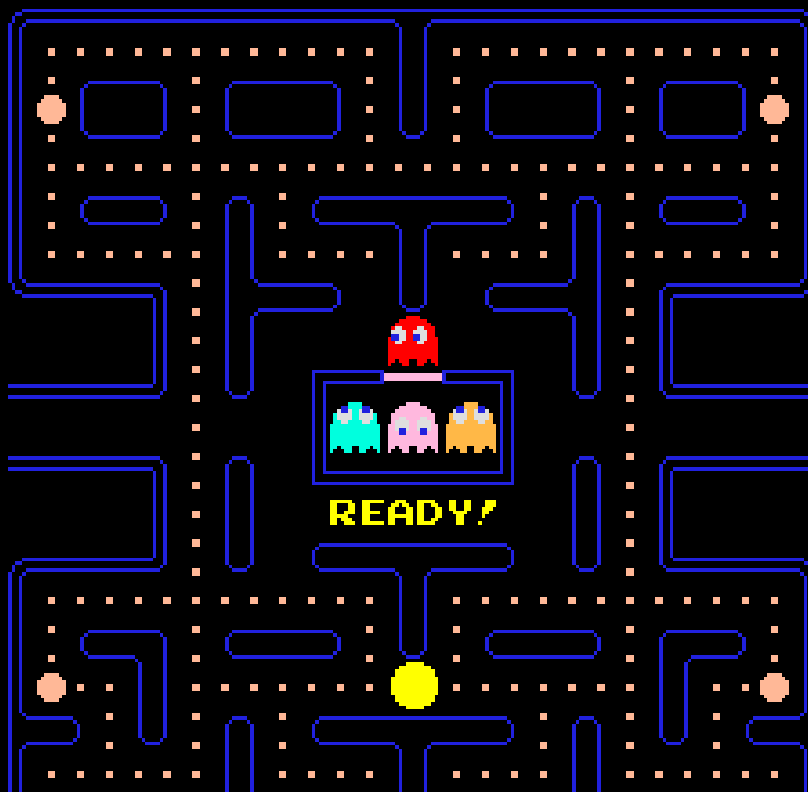


1UP 00

HIGH SCORE 00

MARIO 3230

TOP 12000





AMMO	100%	<table border="1"><tr><td>2</td><td>1</td><td>1</td></tr><tr><td>5</td><td>5</td><td>2</td></tr></table>	2	1	1	5	5	2		0%	ARMOR	<table border="1"><tr><td>BULL</td><td>23</td><td>200</td></tr><tr><td>SHEL</td><td>8</td><td>20</td></tr><tr><td>RUN</td><td>0</td><td>50</td></tr><tr><td>CELL</td><td>0</td><td>300</td></tr></table>	BULL	23	200	SHEL	8	20	RUN	0	50	CELL	0	300
2	1	1																						
5	5	2																						
BULL	23	200																						
SHEL	8	20																						
RUN	0	50																						
CELL	0	300																						

DESIGN RESTRICTIONS – WHAT MAKES A GAME

To be eligible for this course, the game must have following features:

- The game is compiled as a “stand-alone” product and does not need debugging or developer mode, or installation of the toolkit to function
- The game needs to have at least
 - 1 implemented feature from Groups 1-6 each (Levels, Player character, Enemy characters, Sound, UI and controls and Measuring progressin)
 - At least 10 points total from all features.



DESIGN RESTRICTIONS 2 – TECHNICALITIES

There are some limitations on what is accepted as the personal project:

- Self-made (cooperation should be reported in the documents or in-game credits), no teams.
- Bought assets (for example Unity asset store) stuff is allowed. Feel free to use the free stuff, or even buy assets if you feel like it.
 - However, all external assets, including code and models from the Internet sources HAS TO BE referenced.
- Cannot be earlier project, has to be made for this particular course.
- Cannot include illegally acquired assets (everything must be self made, public domain, open source or have license.)
- Compiled and deployed version of the game works with a Windows - workstation with a keyboard and mouse.
 - ..Or with a normal Firefox or Chrome installation.

COURSE SCHEDULE

- This course is now Anytime-course. You have 100 days from your first submitted work (or activation of course registration key) to complete the course.
- **Do the course exam as the last thing you do for the course.** Teaching staff gets an alert on submitted exam, and starts the review process of your works.

There is one deadline, though:

Everything needs to be submitted before 1.8.2023 00:00
because this is when the course Moodle area expires.

GRADING

The course gives 6 study credits with 1-5 grading. The grade is based on two aspects: Project 2 and Course Exam, of which average grade will be the final course grade. To be eligible for final grade, both Project 2 and exam have to be passed with at least grade 1.

Project 2 is graded by implemented feature points as follows:

Minimum requirements met +

10-12 points	= 1
13-14 points	= 2
15-16 points	= 3
17-18 points	= 4
19+ points	= 5

Exam has maximum points of 40 points, so the grading goes

0-19: 0, 20-23: 1, 24-27: 2, 28-31: 3, 32-35: 4, 36+: 5.

Since you can get partial points, and minus points, in the exam the exam points are not rounded up or down for the final grade: 35.95 is still grade 4.

SUMMARY

So in summary, what to do next?



IF NO IDEA ON WHAT TO DO, USE UNITY

Free to use version (including commercial use) available at www.unity3d.com

- Basic tools to lighting
- Physics
- 3D model management
- Scenes (Areas / Levels / Intermission etc.)
- UI overlay
- Particles, particle effects
- Uses C# as main programming language.
- User interface based on the drag-and-drop and attribute management from Inspector
 - Although most of the actual functions defined in scripts which are written like "normal" code.
- Start with the tutorial projects, see where it goes from there.

HOW ABOUT SOMETHING ELSE?

- You do not have to make your own game with Unity.
- If you are already familiar with something else, or own license to, something like Unreal engine or GameMaker Studio feel free to use that software.
 - The only thing that matters is your end-product and that it follows the course restrictions.



SO WHAT TO DO NEXT?

1. **Get yourself a workstation, and set up your work environment (Unity if unsure about else). Watch the lecture videos (there are only 5 of them left since this is almost over)**
2. **Select your tools and design your game**
 - Do a couple of tutorials to get familiar with your tools, design your own game, and present your setup with an online video, and by submitting the GDD via Moodle.
3. **Make your game**
 - After finalizing your own game, your teaser trailer, the source code and playable copy of the game available online. Submit everything via Moodle.
4. **Complete the course exam**
 - If you want to, you can also add your complete game and teaser trailer to the “arcade” are of the course Moodle pages for comments from other course participants and as an inspiration for later participants. *This step is voluntary, and does not affect the course grading.*

CT60A5401 GAME DEVELOPMENT PROJECT



Comments, feedback, want to give your honest opinion?

Email jussi.kasurinen@lut.fi